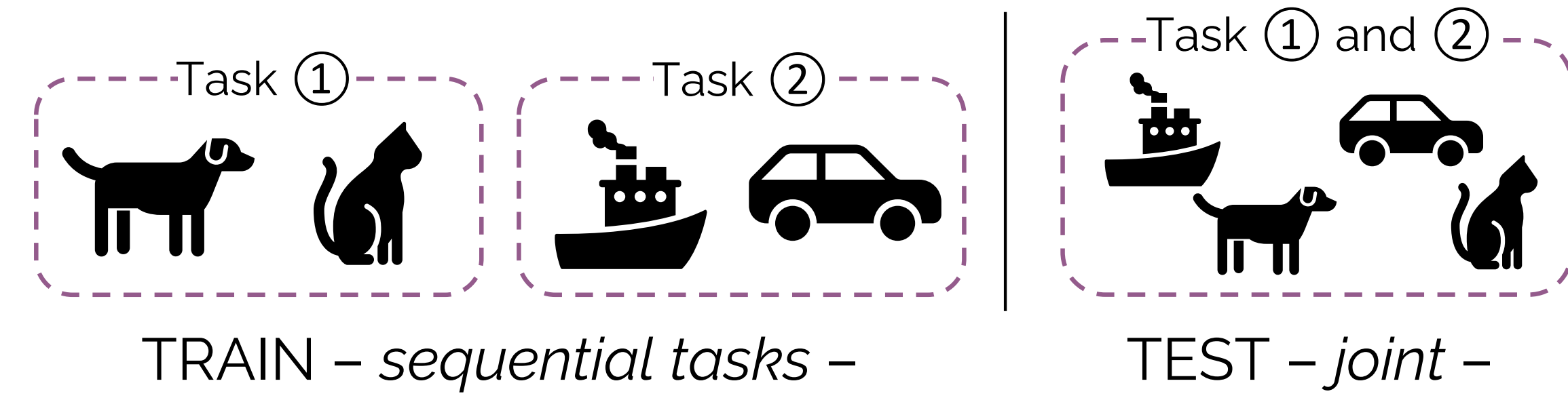


Continual Learning (CL)

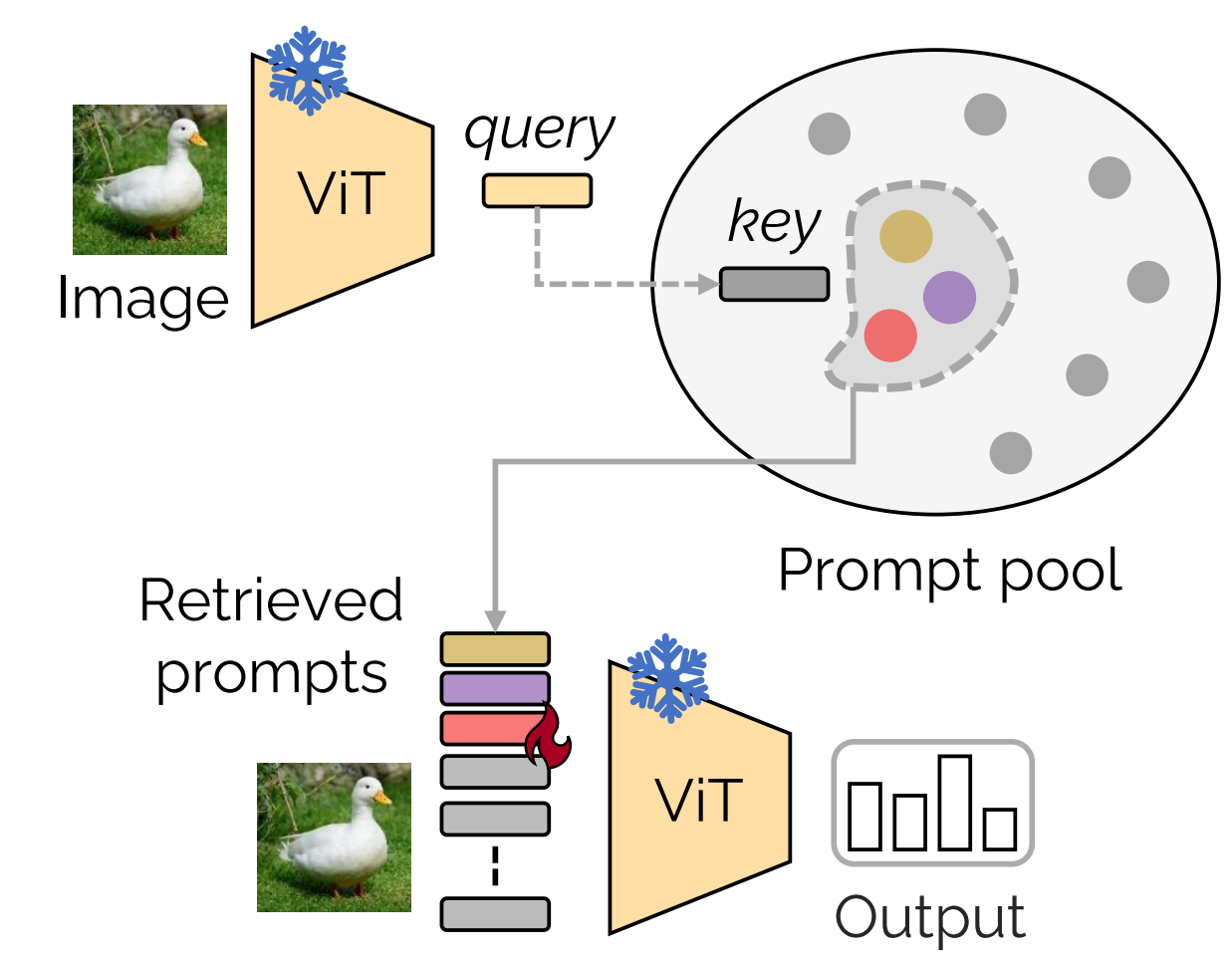
Class-Incremental Learning. Adapting AI models to learn new classes over time (*plasticity*) without forgetting previously learned knowledge (*stability*).



Motivations

Most CL approaches fine-tune **pre-trained** models.

Prompting methods



- 1 A frozen pre-trained model computes the *query*.
- 2 The *query* is compared against a set of pairs <key, prompt> (*prompt pool*).
- 3 The prompt with the highest query-key similarity is used to adjust the model.

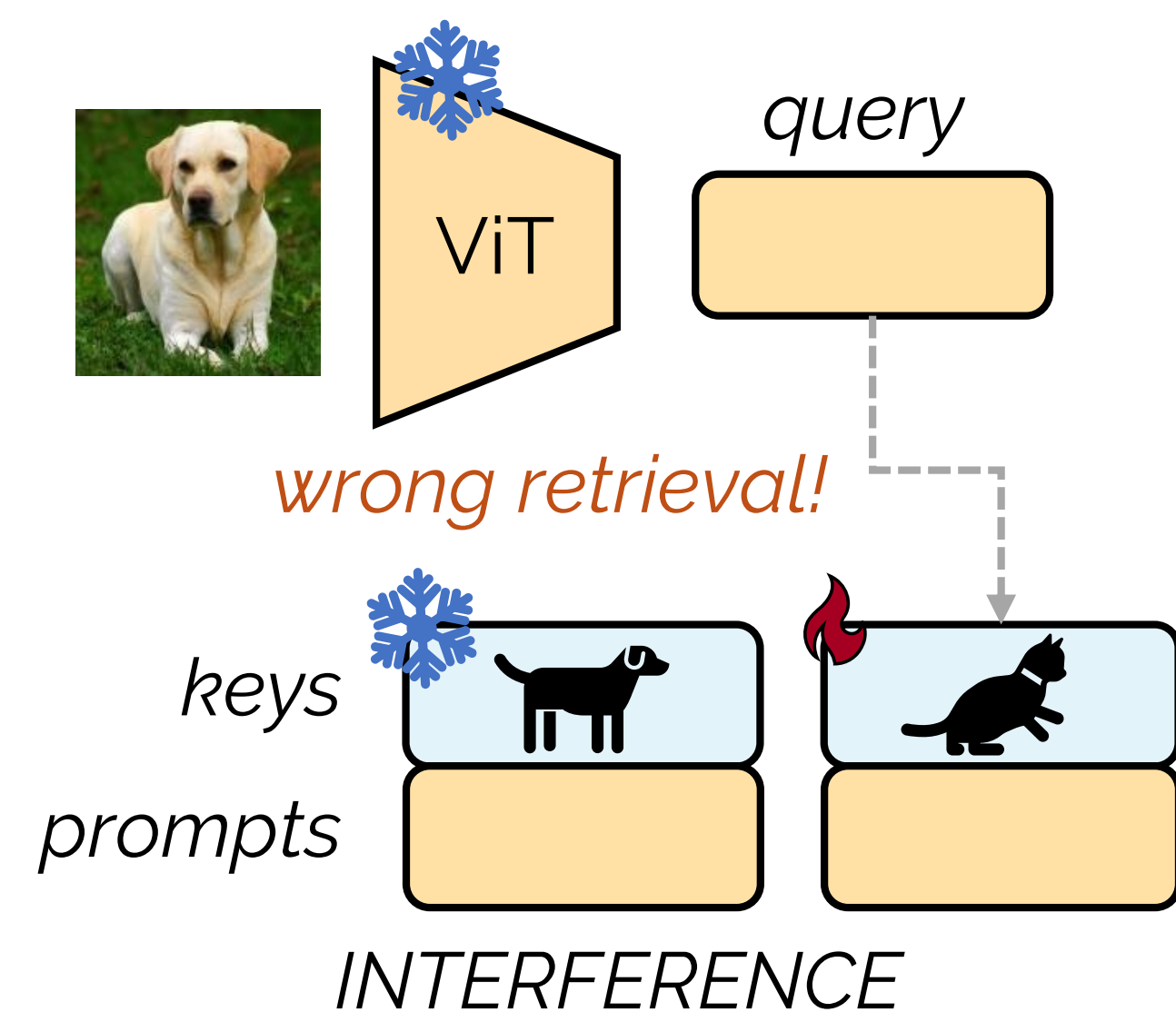


Problem. As keys are learned while tasks progress, the query-key strategy used to retrieve the prompts is also subject to forgetting.

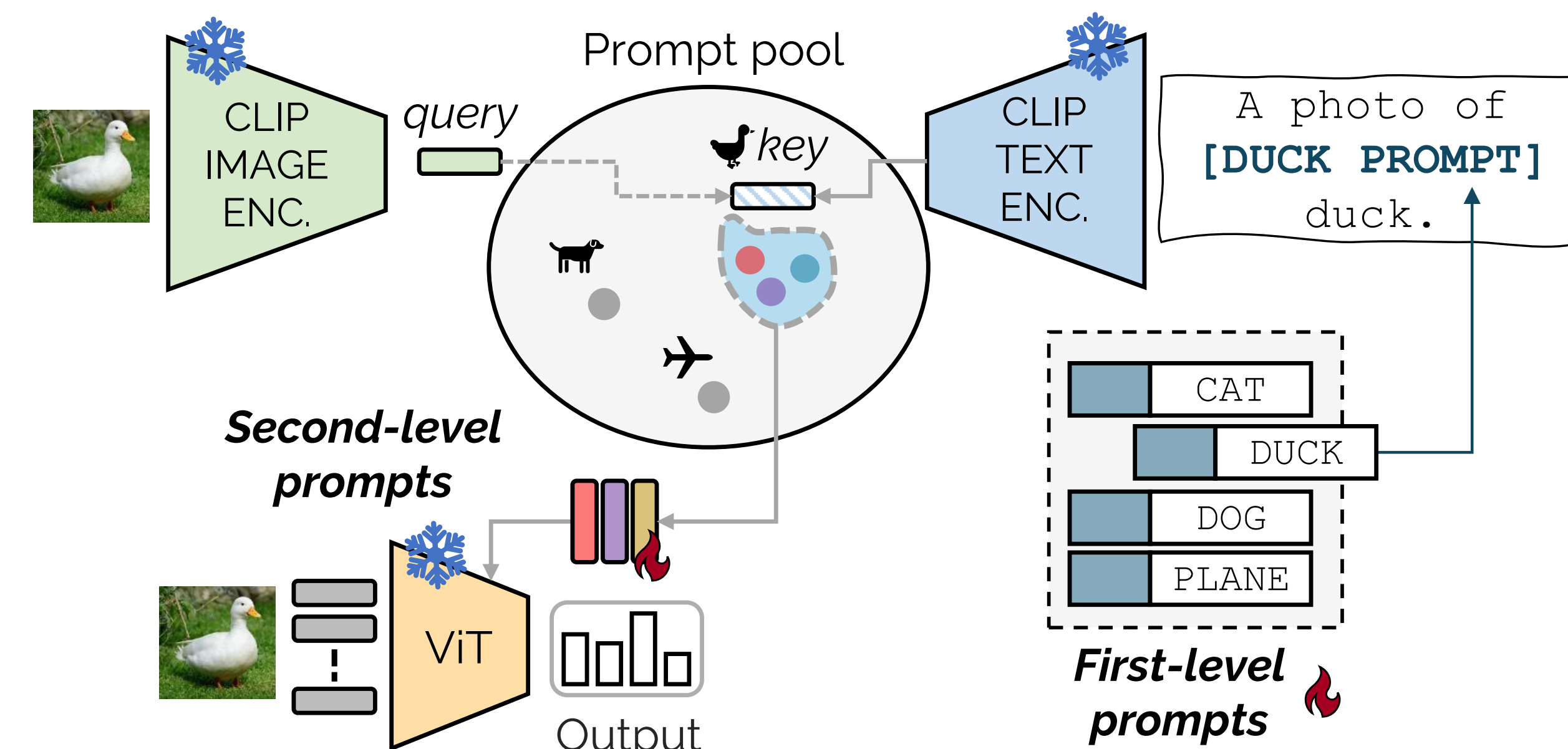
↳ the prompts introduced for new tasks **interfere** with existing ones.



Objective. Enhance *stability* of the strategy used to retrieve prompts.



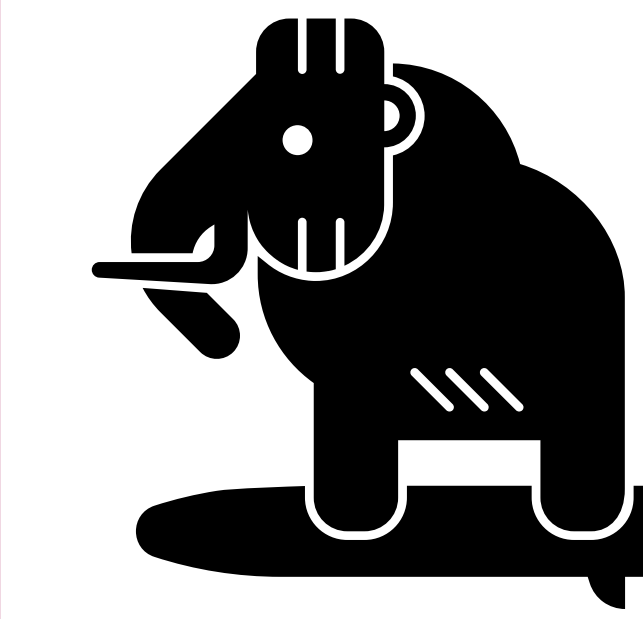
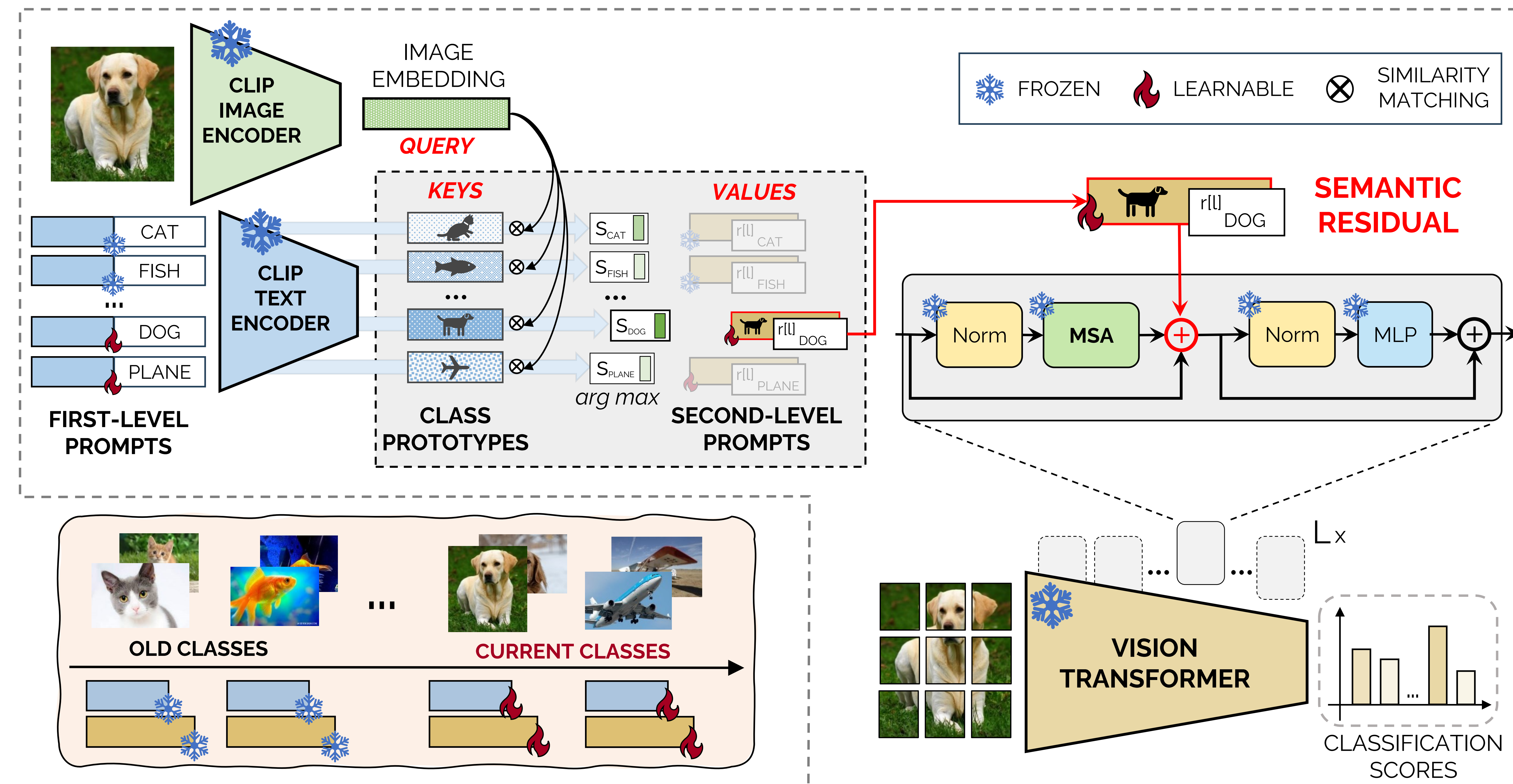
IDEA. To enhance the stability of the selection strategy, we incrementally fine-tune a foundational model (CLIP) and use it to select our prompts.



Our approach: STAR-Prompt

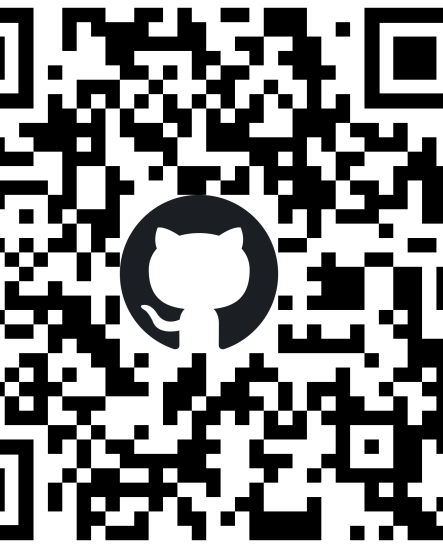
It relies on a **two-level** prompting pipeline and a **generative replay** approach to create stable class prototypes.

- 1 **First stage.** The CLIP text encoder is trained with class-level prompts. The corresponding textual embeddings serve as keys of the second prompting level.
- 2 **Second stage.** These keys are matched against a query produced by the CLIP visual query. The corresponding prompt is used to fine-tune an ImageNet pre-trained ViT.



Mammoth

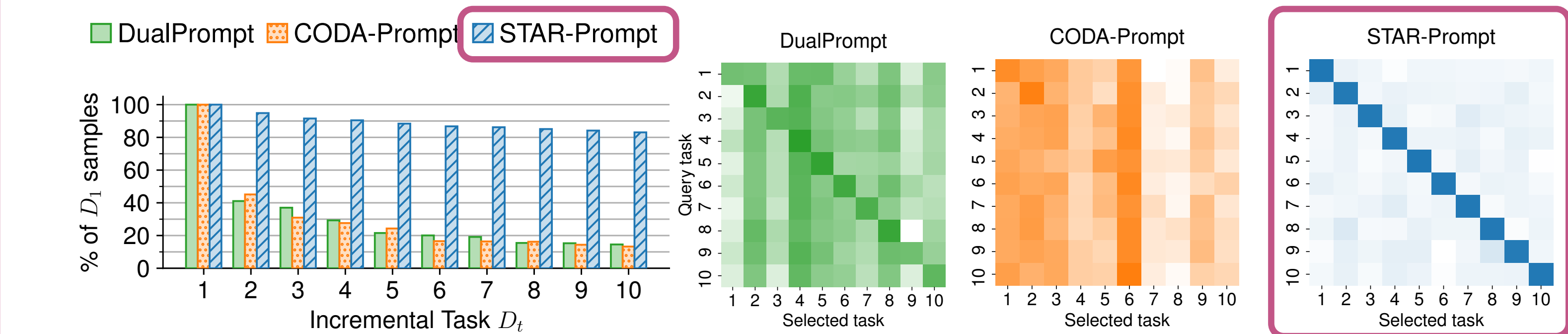
A PyTorch benchmark library for Continual Learning with 20+ datasets and 50+ methods.



Experiments



Stability. Compared to existing approaches, our query-key selection based on CLIP retrieves the set of prompts **relevant** to a given task, and not those introduced for subsequent tasks.



- 1 **Insight.** The more precise the prompt selection, the more resilient the resulting model is to *catastrophic forgetting*.
- 2 **Plasticity.** CLIP alone is not sufficient; a dedicated plasticity mechanism like our second prompting stage is crucial.

Model	ImgNet-R	Cars-196	EuroSAT	ISIC	Avg.
Joint (STAR-Prompt)	90.03	89.00	97.21	78.25	88.62
Joint (ViT-B/16)	79.60	80.31	98.19	88.31	86.60
Zero-shot CLIP	85.36	76.46	54.50	29.14	61.37
DER++	56.66	53.66	93.08	65.68	67.27
L2P	66.49	38.18	46.34	47.13	49.53
DualPrompt	68.50	40.14	71.39	49.99	57.51
CODA-Prompt	75.45	31.99	63.12	44.87	53.86
AttriCLIP	86.25	70.98	57.51	26.77	60.38
SLCA	77.00	67.73	88.69	59.19	73.15
STAR-Prompt	89.83	87.62	93.70	66.67	84.46
Ablations					
Classify with first-level keys w_c	88.16	87.57	90.12	58.87	81.18
w/o first-level prompts	88.97	79.88	86.25	58.68	78.44
w/o Generative Replay	88.55	87.17	83.78	50.76	77.57